
UNIT 8 MODEL SPECIFICATION ISSUES*

Structure

- 8.0 Objectives
- 8.1 Introduction
- 8.2 Possible Problems in Specification
 - 8.2.1 Importance of Model Specification
 - 8.2.2 Sources Specification Error
 - 8.2.3 Implications of Specification Error
- 8.3 Inclusion of Variables a Model
 - 8.3.1 Inclusion of Irrelevant Variables
 - 8.3.2 Omission of Relevant Variables
- 8.4 Specification Error Test
- 8.5 Model Selection Criteria
 - 8.5.1 The R^2 and Adjusted- R^2 Criteria
 - 8.5.2 The AIC and SIC Criteria
 - 8.5.3 Mallows's C_p Criterion
- 8.6 Caution about Model Selection Criteria
- 8.7 Let Us Sum Up
- 8.8 Key Words
- 8.9 Some Useful Books
- 8.10 Answers/ Hints to Check Your Progress Exercises

8.0 OBJECTIVES

After going through this unit you should be in a position to:

- explain the nature of problems associated with model specification;
- identify the consequences of model misspecification;
- find out the consequences of including an irrelevant variable;
- find out the consequences of excluding a relevant variable;
- detect a mis-specified model and suggest remedial measures; and
- explain various tests for model selection.

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8.1 INTRODUCTION

Regression analysis, as you know, is based on certain assumptions. These assumptions pertain to specification of the functional form and behaviour of the error variable. The inferences drawn from OLS estimation, therefore, are based on fulfilment of these assumptions.

While estimating a regression model, correct specification of the model is very important for obtaining unbiased and efficient estimates. However, several questions arise: How do we detect whether the OLS assumptions are met for a given dataset? Is the selected functional form appropriate for analysis? Are the variables included in the model relevant? In the presence of any such specification issue, what is the impact on the estimators? Are the inferences drawn still valid?

In this unit we will look out for the major specification issues and the impact it will have on prediction.

8.2 POSSIBLE PROBLEMS IN SPECIFICATION

Although extremely significant, specification is least understood aspect of regression analysis. In general, model specification should have theoretical basis rather than empirical or methodological one.

8.2.1 Importance of Model Specification

The causal relationship between regressand (explained variable) and regressor (explanatory variable) in a multiple regression model involves three steps: (i) specification of the model, (ii) estimation of the parameters of this model, and (iii) interpretation of these parameters.

Since specification forms the first and most critical step, any error in it will lead to error in estimation and interpretation. If the regressors in multiple regression analysis are independent and uncorrelated, their partial regression coefficients will be equal to their simple regression coefficients. Consequently, addition or deletion of orthogonal independent variables will not affect the partial regression coefficient of other independent variables in a multiple regression model. But the standard error and hence error variance of dependent variables are significantly affected by inclusion or omission of orthogonal independent variables. Inclusion of independent variables in regression equation increases the standard error of the partial regression coefficient by decreasing their degrees of freedom. Simultaneously, however, the addition of an independent variable may decrease these same standard errors by decreasing the error variance in the dependent variable.

8.2.2 Sources of Specification Error

There could be numerous ways in which a bias or an error can enter a regression model. Some of these are (i) inclusion of redundant/ superfluous/ irrelevant variables, (ii) omission of relevant variables, (iii) incorrect functional form, (iv)

incorrect assumption regarding the error term, and (v) error in measurement of variables.

Let us consider the multiple regression model

$$Y = \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + u \quad \dots(8.1)$$

We have two sets of assumptions: (i) specification of the model, and (ii) specification of the error variable. The assumptions regarding specification of the model are as follows:

- a) The regression model is specified correctly – all relevant variables are included, and no relevant variable is excluded from the model.
- b) The model is linear in parameters.
- c) The variables are measured without errors.
- d) The explanatory variables are independent of one another – there is no perfect correlation among them.

The assumptions regarding specification of the error variable are as follows:

- (i) $E(u_i) = 0$
- (ii) $E(u_i)^2 = \sigma^2$
- (iii) $E(u_i u_j) = 0$ for $i \neq j$
- (iv) $E(X_i u_i) = 0$ (regressor is non-stochastic; the error term is not correlated with any of the explanatory variables)

Violation of the above assumptions results in several types of problems in the desirable properties of the estimators.

8.2.3 Implications of Specification Error

Let us begin with assumption (a) given above. A model refers to a simplified version of reality. It allows us to explain, analyse and predict economic behavior. An economic model can be for a microeconomic agent such as household or firm. In macroeconomics, it represents the behavior of the economy. In economic models we identify relevant economic variables (such as income, output, expenditure, investment, savings, exports, etc.) and establish relationship among them. The relationships among these variables may be expressed through diagrams or mathematical equations. There could be economic models without mathematical expressions, but such models may not be precise.

While building an econometric model we first consider the logic or economic theory behind the model. The empirical or methodological considerations come later. The accuracy of the estimated parameters and the inferences drawn from the model depend upon the correct specification of the model.

An econometric model comprises a dependant variable, independent variable(s), and an error term. As you know, regression signifies a cause-and-effect relationship – variation in the dependent variable is explained by the variation in

the independent variables. Thus, while specifying a regression model, we should see to it that the dependant variable is logically explained by the independent variables. If an independent variable does not explain the dependent variable, it should not be included in the model.

Next is the functional form of the regression model, which should be specified correctly. Let me illustrate the point through an example. In the case of a firm, we assume that there are two factors of production, viz., capital and labour. We aggregate all types of labour into a homogeneous category – we do not distinguish between a manager and a worker in the field! Thus, we ignore the details and concentrate on the major issues in a model. Secondly, we assume that the production function takes a particular form, say Cobb-Douglas. But it is just an assumption! The production function, in reality, could be of some other form also such as CES or translog. Thus, we need to logically explain the functional form (regression equation) of the model.

Assumption (b) says that the regression model is linear in parameters. Equation (8.1) is linear in parameters (there are no such terms as β_i^2 , for example). Suppose our regression model is

$$Y_i = \beta_1 + \beta_2 X_i + \beta_3 X_i^2 + u_i \quad \dots (8.2)$$

Equation (8.2) is a quadratic function, but it is still linear in parameters. Examples of non-linear regression models are logarithmic functions, logistic functions, trigonometric functions, exponential functions, etc. Some of these functions can be transformed to linear functions (see Unit 3). If we cannot transform a non-linear function into a linear function, we cannot estimate the non-linear function by OLS method.

Many a time we specify the functional form correctly, but the variables are not measured correctly. There could be errors in measurement of the independent variable(s) or in the dependent variable. Such issues are called error-in-variables; and we will deal with them in Unit 12. Remember that error in variable is different from the error term (u_i) included in a regression model. The error term reflects the variables that could not be included in the regression equation (excluded variables). Measurement error in a variable, on the other hand, reflects the situation where the variable is included in the model, but it is not correctly measured. There could be several sources of measurement error: (i) it may not be possible to measure certain variables (e.g. taste, preference, business environment, intelligence, etc.), or (ii) the variable can be defined conceptually but it may not be possible to measure the variable correctly (e.g., education – it can be measured by a proxy, that is, years of schooling). There are certain other factors that lead to errors in variables such as: (i) non-response error, (ii) reporting error, and (iii) computing error.

In case the assumption (d) is violated we have the problem of multicollinearity. It is a problem observed in multiple regression models, when we include too many

variables in a regression equation. It is basically a data problem. Independence of the explanatory variables is ensured when the matrix X has full rank. We will discuss problems arising out of multicollinearity in Unit 10.

8.3 INCLUSION OF VARIABLES IN A MODEL

Regression analysis derives its robustness from the assumption that the econometric model under study is correctly specified. We discuss below the consequences of (i) inclusion of irrelevant variables, and (ii) omission of relevant variables in a regression model.

Each of the above problem results in a different kind of bias.

8.3.1 Inclusion of Irrelevant Variables

Let us consider the case where certain irrelevant variable is included in the regression model. Suppose the true model is

$$Y_i = \beta_0 + \beta_1 X_{1i} + u_i \quad \dots (8.3)$$

But we somehow include a redundant variable, i.e., we estimate the following equation:

$$Y_i = \beta_{0s} + \beta_{1s} X_{1i} + \beta_{2s} X_{2i} + v_i \quad \dots (8.4)$$

For the true model (8.3), the slope coefficient is expressed as

$$\hat{\beta}_1 = \frac{\sum yx_1}{\sum x_1^2} \quad \dots (8.5)$$

which is unbiased.

For the model (8.4) that we have taken, we obtain

$$\tilde{\beta}_1 = \hat{\beta}_{1s} = \frac{(\sum yx_1)(\sum x_2^2) - (\sum yx_2)(\sum x_1x_2)}{\sum x_1^2 \sum x_2^2 - (\sum x_1x_2)^2} \quad \dots (8.6)$$

Now the true model in deviation form is

$$y_i = \beta_1 x_1 + (u_i - \bar{u}) \quad \dots (8.7)$$

In equation (8.6) let us substitute the value of y_i from (8.7) and simplify. If we take expected value of (8.6), after substitution and simplification, we obtain

$$E(\tilde{\beta}_1) = E(\hat{\beta}_{1s}) = \beta_1 \frac{\sum x_1^2 \sum x_2^2 - (\sum x_1x_2)^2}{\sum x_1^2 \sum x_2^2 - (\sum x_1x_2)^2} \quad \dots (8.8)$$

From equation (8.8) we find that

$$E(\tilde{\beta}_1) = \beta_1$$

Thus, inclusion of an irrelevant variable provides us with unbiased estimator of β_1 .

The estimator of the redundant variable $\hat{\beta}_{2s}$ is given by

$$\hat{\beta}_{2s} = \frac{(\sum yx_2)(\sum x_1^2) - (\sum yx_1)(\sum x_1x_2)}{\sum x_1^2 \sum x_2^2 - (\sum x_1x_2)^2} \quad \dots (8.9)$$

If we substitute for y_i from (8.7) in (8.9) and re-arrange terms, we obtain

$$E(\tilde{\beta}_2) = E(\hat{\beta}_{2s}) = \beta_2 \frac{(\sum x_1 x_2)(\sum x_1^2) - (\sum x_1 x_2)(\sum x_1^2)}{\sum x_1^2 \sum x_2^2 - (\sum x_1 x_2)^2} \dots (8.10)$$

$$\text{Thus, } E(\tilde{\beta}_2) = E(\hat{\beta}_{2s}) = 0$$

So, we find that $\hat{\beta}_{2s}$, which is absent from the true model, has its coefficient '0'.

Thus, we obtain unbiased estimators for both the parameters. This leads us to conclude that inclusion of irrelevant variables is not that harmful as omission of relevant variables. As an extra variable is added to the model, we observe that there is an increase in R-squared. The variance of the parameters will not be efficient.

Therefore, the specification error in the case of inclusion of irrelevant variables in the model will produce unbiased but inefficient least squares estimators of the parameters. The larger variance reduces the precision of the estimates resulting in wider confidence intervals. This may lead to type II error (the error of not rejecting a null hypothesis when the null hypothesis is actually not true).

8.3.2 Omission of Relevant Variables

Now let us look into the other side of the spectrum – excluding a relevant variable. Since a relevant variable is not included in the model (although it influences the dependent variable) its impact will be included in the residuals. As a result, the residuals will show a systematic pattern rather than being white noise as required by Gauss-Markov theorem. Also, the coefficient of the included variable will be biased.

Suppose the true equation (in deviation form) is

$$y = \beta_1 x_1 + \beta_2 x_2 + u \dots (8.11)$$

Instead of estimating equation (8.11) suppose we omitted x_2 . The following equation is estimated,

$$y = \beta_1^* x_1 + e \dots (8.12)$$

Equation (8.12) is a case of omitted variable, and hence incorrect model specification. In the model with omitted variable (incorrect model) the estimate of β_1^* is

$$\hat{\beta}_1^* = \frac{\sum x_1 y}{\sum x_1^2} \dots (8.13)$$

In order to calculate the bias in the estimated value of β_1 in the incorrect model (equation (8.12)) as compared to the true model (equation (8.11)), we take the following steps:

We substitute the expression of y from the true model (8.11) in equation (8.13).

This gives us

$$\hat{\beta}_1^* = \frac{\sum x_1 (\beta_1 x_1 + \beta_2 x_2 + u)}{\sum x_1^2} = \beta_1 + \beta_2 \frac{\sum x_1 x_2}{\sum x_1^2} + \frac{\sum x_1 u}{\sum x_1^2} \dots (8.14)$$

Since $E(\sum x_1 u) = 0$ we get

$$E(\hat{\beta}_1^*) = \beta_1 + \beta_2 b_{21} \quad \dots (8.15)$$

where $b_{21} = \frac{\sum x_1 x_2}{\sum x_1^2}$ is the regression coefficient from a regression of x_2 (omitted variable) on x_1 .

Thus, $\hat{\beta}_1^*$ is a biased estimator for β_1 and the bias is given by

$$E(\hat{\beta}_1^*) - \beta_1 = \beta_2 b_{21} \quad \dots (8.16)$$

Thus, bias = (coefficient of the excluded variable, β_2) $\times$ (regression coefficient in a regression of the excluded variable on the included variable, b_{21})

The variance of $\hat{\beta}_1^*$ (parameter of the incorrect model) can also be derived by using the formula for variance. As it is a bit complex, we do not present it here. You should note that the variance of $\hat{\beta}_1^*$ is higher than that of $\hat{\beta}_1$. An implication of the above is that usual tests of significance concerning parameters are invalid, if some of the relevant variables are excluded from a model.

In order to detect omitted variables, residuals ($\hat{u}_i$) can be examined for any pattern that they exhibit (see Unit 5). If a variable is omitted, it will be reflected in the form of a systematic pattern in the residuals. If the residuals show autocorrelation which can be detected using Durbin-Watson d-statistic (see Unit 9), there exists model misspecification where some variable is omitted from the model.

Thus, we know that

- (i) When an irrelevant variable is included in the model: (a) the estimators of parameters are unbiased, (b) there is a decline in the efficiency of the estimators, and (c) the estimator of the error variance is unbiased. Thus, conventional tests of hypothesis are valid. The inferences drawn could be somewhat erroneous.
- (ii) When a relevant variable is dropped from the model: (a) estimators of parameters are biased, (b) decline in the efficiency of the estimators, and (c) estimator of error variance is biased (over-estimate). Thus, conventional tests of hypothesis are invalid. The inferences drawn are faulty.

Recall from Unit 3 that there two approaches to building an economic model: (i) general to specific (also called top-down approach), and (ii) specific to general (also called bottom-up approach). In the first approach we consider a general model by including many variables. Subsequently, we exclude the irrelevant variables. In the second approach, we begin with a simple model and keep on adding variables based on our logic.

Although, there is no consensus on the approach to be followed in econometric model building, researchers prefer the general-to-specific approach. Now you can understand the reason behind it – if we exclude certain relevant variable(s) from a

model, we do not obtain unbiased estimators even for the variables included in the model!

When we follow the general-to-specific approach, we may retain only those variables which are statistically significant (i.e., we exclude the variables which are not statistically significant). In the case of the specific-to-general approach, i.e., expanding the model after testing for the statistical significance of the variables is criticized because the true level of significance and the nominal level of significance differ if all the variables are not included in the model.

Check Your Progress 1

- 1) List the assumptions of the classical regression model. What problems do you encounter if these assumptions are violated?

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- 2) Do you agree that correct specification of an econometric model is important? Why?

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- 3) List three types of specification error that we encounter in an econometric model.

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- 4) Explain the consequences of inclusion of an irrelevant variable.

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- 4) Explain the consequences of excluding a relevant variable.

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8.4 SPECIFICATION ERROR TEST

There is a tendency on the part of researchers to assume a linear relationship between variables. This however is not always true. If the true relationship is non-linear and we take a linear regression model for estimation, we will not be able to draw correct inferences. Hence, detection of the specification error can help us in obtaining efficient estimates. Once it is found that specification errors are made, finding a remedy is not a major problem.

One general test for model specification, that is very popular, is the RESET (Regression Specification Error Test). The idea underlying RESET is simple.

Let us assume that we consider the model (8.17) as given below for estimation.

$$y = \beta_1 x_1 + \beta_2 x_2 + u \quad \dots (8.17)$$

The RESET introduces quadratic and cubic functions of the estimated dependant variable in the regression model and compares it to the model without the additional regressors using F-test. If the increase in R^2 is statistically significant, there exists model misspecification. Now we will discuss the step-by-step RESET,

- (i) From the chosen model $y = \beta_1 x_1 + \beta_2 x_2 + u$, obtain the estimated y , that is $\hat{y}$.
- (ii) Re-run the above chosen model introducing $\hat{y}$ in some form as an additional regressor (quadratic and cubic form is quite popular and yields good results). Thus, the new model will be

$$y = \beta_1 x_1 + \beta_2 x_2 + \beta_3 \hat{y}^2 + \beta_4 \hat{y}^3 + v \quad \dots (8.18)$$

- (iii) Let the R^2 obtained from the model in the second step above be R_{new}^2 and that obtained from step one be R_{old}^2 . The F test can be used to find out if the increase in R^2 from using the model in step two is statistically significant using the following F-test from Unit 4:

$$F = \frac{(R_{new}^2 - R_{old}^2) / \text{number of new regressors}}{(1 - R_{new}^2) / (n - \text{number of new parameters in the model})} \quad \dots (8.19)$$

- (iv) If the computed F-value is statistically significant, say, at the 5 percent level, we reject the null hypothesis that the model is correctly specified. It

implies we accept the alternate hypothesis that the regression model is mis-specified.

Limitations of the RESET

The RESET does not test for omitted variables. It tests whether the functional form of the model is correctly specified or not. If there is non-linearity in the functional form and we consider a linear model, the RESET test will reject the null hypothesis. In other words, we accept the alternate hypothesis. The alternative hypothesis in the RESET, however, is that there is misspecification of the functional form. Thus, an important limitation of the RESET is that it does not provide further leads: What should we do if there is misspecification? What functional form should we adopt?

8.5 MODEL SELECTION CRITERIA

From the discussion so far, we observe that there could be many competing econometric models for certain economic phenomenon. For example, for quarterly data on GDP of India, there could be several autoregressive-moving average (ARMA) models before us. There could be certain seasonal variation in the GDP data along with the secular trend. Such variations are represented by ARMA models with different orders of AR and MA. Similarly, for the estimation of production function of a firm, there are quite a few functional forms such as Cobb-Douglas, CES, translog, etc. A pertinent issue is selection of regressors in a model.

There are certain criteria on the basis of which we select the most appropriate regression model for particular situation. We discuss these criteria below.

8.5.1 The R^2 and Adjusted- R^2 Criteria

We have discussed the concept of coefficient of determination (R^2) in Unit 4. As you know, the coefficient of determination indicates the explanatory power of a model. If, for example, $R^2 = 0.82$ we can infer that 82 per cent variation in the dependent variable is explained by the explanatory variable in the model. The R^2 of a model measures the goodness of fit of a regression model and lies between 0 and 1.

$$R^2 = \frac{\text{explained sum of squares}}{\text{total sum of squares}} = 1 - \frac{\text{residual sum of squares}}{\text{total sum of squares}} \quad \dots (8.20)$$

A higher value of R^2 indicates better fit of a model and vice versa. The R^2 measures only in-sample goodness of fit and it cannot forecast the out-of-sample observations, which happen to be an important function of a regression model. Second, we can compare among models having the same dependant variable. Third, with every additional regressor, even if irrelevant, the R^2 -value does not decrease. Hence a researcher may go on adding more independent variables although this will increase the variance of forecast error.